StockTrader: Signals, Strategies, and Disagreement

Comparative Analysis Report

Momentum Trader vs. Value Contrarian

April 14, 2026

Agentic Technology | Coding Assignment 2

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1. Strategy Selection and Rationale

This assignment pairs two strategies with fundamentally opposed behavioral philosophies: the Momentum Trader and the Value Contrarian. These two were chosen deliberately because they interpret identical market data in diametrically opposite ways under the most common market condition — a rising stock.

The Momentum Trader operates on the principle that price trends persist. It buys stocks that are already rising, confirmed by price sitting above its 20-day and 50-day moving averages, and supported by volume. It treats a high RSI as evidence of strength, not danger. The strategy's core belief is that winners keep winning in the short to medium term.

The Value Contrarian operates on the opposite premise: markets systematically overreact, creating buying opportunities in fear and selling opportunities in greed. It ignores moving averages entirely and instead monitors distance from 52-week highs, RSI levels, and recent drawdowns. An RSI above 70 combined with a price near the 52-week high is its sell signal; a deep drawdown with a low RSI is its buy signal.

Expected disagreement: The sharpest conflicts were anticipated on high-momentum stocks like NVDA and INTC, where strong uptrends would trigger a Momentum BUY at the same moment that elevated RSI and proximity to 52-week highs trigger a Contrarian SELL. Sideways or declining stocks were expected to produce more nuanced splits. This prediction proved accurate across the six stocks analyzed.

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2. System Architecture

The system is implemented in pure Python using the standard library's concurrent.futures module for parallelism. No external agent framework is required. The pipeline has four stages:

```
                          START  (ticker input)
                                      |
       [ Market Data Component ]   ←  yfinance only, zero LLM calls
                          |                     |
    [ Strategy A ]      [ Strategy B ]   ←  concurrent.futures parallel
                                   threads
                   Momentum Trader     Value Contrarian
                          |                     |
                          +--------+-----------+
                          [ Evaluator Component ]
                          |                     |
                [ Consensus ]    [ Disagreement Analysis ]
                                      |
                            END  (save to JSON)
```

Each component maps to a dedicated Python module: market_data.py handles all yfinance calls and indicator computation; strategies.py contains both agent functions; evaluator.py handles comparison and LLM synthesis; orchestration.py wires them together and saves JSON output.

The parallelism guarantee is critical: both Strategy A and Strategy B are submitted simultaneously to a ThreadPoolExecutor with max_workers=2. Neither agent sees the other's output before producing its own recommendation. Both receive the same formatted market data string derived from the same dictionary.

The LLM is Ollama running locally with the llama3.2 model. All prompts are stored as plain text files in the prompts/ directory and loaded at runtime, making them easy to inspect and modify without touching Python code.

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3. Stock Selection and Rationale

Six stocks were selected to represent a range of market conditions. The goal was to create scenarios where the two strategies would be forced to reach different conclusions — and one scenario where they would converge.

Ticker	Market Condition	Rationale	Key Data
MSFT	Steady large-cap in recovery	Down 27% from 52w high but recovering week-over-week. Tests whether momentum signals early recovery.	RSI 57.8, -27% from 52w high, +5.3% 7d
NVDA	High-momentum growth	Near 52w high with strong momentum. Expected maximum disagreement between strategies.	RSI 68, -5% from 52w high, +10.8% 7d

PFE	Declining large-cap	Below both MAs with negative momentum. Tests SELL vs. contrarian value opportunity.	RSI 48, below both MAs, momentum -2.04
INTC	Explosive momentum surge	+40% in 30 days, RSI 74. Designed to be the clearest possible disagreement case.	RSI 74, +40.2% 30d, momentum +34.81
GME	Sideways / range-bound	Flat 30-day return but 31% below 52w high. Expected to test whether strategies converge.	RSI 57.9, -31% from 52w high, -0.7% 30d
TSLA	Recent significant decline	Down 26% from high, below both MAs. Tests whether Contrarian buys the fear while Momentum sells.	RSI 46, -25.7% from 52w high, -9.7% 30d

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4. Results by Stock

4.1 Summary Table

Ticker	Momentum	Contrarian	Confidence A	Confidence B	Agreement
MSFT	**BUY**	**HOLD**	8/10	8/10	**DISAGREE**
NVDA	**BUY**	**SELL**	9/10	8/10	**DISAGREE**
PFE	**SELL**	**BUY**	9/10	8/10	**DISAGREE**
INTC	**BUY**	**SELL**	9/10	8/10	**DISAGREE**
GME	**BUY**	**BUY**	8/10	8/10	**AGREE**
TSLA	**SELL**	**BUY**	8/10	8/10	**DISAGREE**

Final: 1 agreement, 5 disagreements across 6 stocks.

4.2 Per-Stock Analysis

MSFT — BUY vs. HOLD

Microsoft presents a recovery scenario: price is above both moving averages and up 5.3% in 7 days, but still 27% below its 52-week high. The Momentum Trader correctly identified the short-term upward trend and issued a BUY at confidence 8. The Value Contrarian, applying its thresholds, reached HOLD: RSI of 57.8 falls in the neutral range (40–65) and the 27% drawdown, while significant, does not reach the deep-fear territory that triggers a contrarian buy. Notably, the evaluator misread the result in its analysis text, calling it a convergence — this is a documentation error in the LLM output and is examined in the Failure section.

Excerpt from MSFT.json — Momentum Trader justification:

> *MSFT's price is above both the 20-day MA ($376.22) and the 50-day MA ($391.94), indicating a bullish trend. The recent 30-day change of -1.36% is positive, and the momentum score of +1.29 suggests a slight uptrend.*

Excerpt from MSFT.json — Value Contrarian justification:

> *RSI is neutral at 57.8, and the stock is within 10% of its 52-week high, but not oversold. The current drawdown is -27.2%, which is below the threshold for a buy signal.*

NVDA — BUY vs. SELL

NVDA is the clearest high-momentum disagreement. At $196.51, the stock is above both moving averages, up 10.78% in 7 days, and carries a momentum score of +8.93 — a strong BUY for the Momentum Trader at confidence 9. For the Value Contrarian, the same data triggers a SELL: RSI at 68.07 exceeds the 65 threshold, and the stock sits only 5% below its 52-week high, satisfying both sell conditions simultaneously.

Excerpt from NVDA.json — Value Contrarian justification:

> *NVDA's RSI of 68.07 is above 65, and it's within 10% of its 52-week high, indicating overbought conditions. Additionally, the stock has surged more than 15% in the past 30 days (+7.69%) and is currently above its 20-day MA, suggesting a potential sell signal.*

PFE — SELL vs. BUY (opposite directions)

PFE produces the sharpest reversal: the Momentum Trader issues a SELL at confidence 9 while the Value Contrarian issues a BUY at confidence 8. Price sits below both moving averages with a negative momentum score of -2.04 and a negative 30-day return, triggering every Momentum SELL condition. The Value Contrarian reads the same data differently: the stock is down from recent highs and showing what the agent interprets as an overreaction selloff. Both agents cite specific numbers and both are internally consistent with their philosophies.

INTC — BUY vs. SELL (maximum divergence)

INTC is the most extreme case in the dataset. The stock surged +40.24% in 30 days, gained +26.66% in a single week, has a momentum score of +34.81, and is only 2.1% below its 52-week high with RSI at 74.02. The Momentum Trader issues a BUY at confidence 9 — this is exactly the explosive uptrend it follows. The Value Contrarian issues a SELL at confidence 8 — RSI above 70 and proximity to the 52-week high simultaneously satisfy its hardest sell triggers.

Excerpt from INTC.json — Momentum Trader justification:

> *The stock is above both the 20-day MA ($50.05) and the 50-day MA ($47.88), indicating a bullish trend. The recent price change (30d) is positive (+40.24%) and the momentum score is +34.81, further supporting the buy signal.*

Excerpt from INTC.json — Value Contrarian justification:

> *INTC is overbought with an RSI of 74.02, which is above the neutral range of 40-65. Additionally, its price is within 10% of its 52-week high, making it a sell signal.*

GME — BUY vs. BUY (the only agreement)

GME is the only stock where both strategies converge. At $24.03, the stock is above both moving averages (satisfying the Momentum BUY condition) and is simultaneously 31.36% below its 52-week high with a max 90-day drawdown of -42.64% (satisfying the Contrarian BUY condition). This is the rare scenario where a stock is both trending upward locally and deeply discounted from its peak — two independent signals pointing the same direction.

TSLA — SELL vs. BUY

TSLA produces a clean and analytically interesting split. With the price below both MAs, a momentum score of -5.42, and a 30-day return of -9.7%, the Momentum Trader correctly identifies a downtrend and issues a SELL at confidence 8. The Value Contrarian reads the -25.66% drawdown from the 52-week high and RSI of 45.88 as genuine fear, issuing a BUY at confidence 8. Both are correct within their frameworks — this is the definitional case where the same data means completely different things depending on your philosophy.

Excerpt from TSLA.json — Value Contrarian justification:

> *TSLA is oversold with an RSI of 45.88, currently 20% below its 52-week high of $489.88. The recent 30-day change of -9.70% indicates a significant decline, suggesting genuine fear. Additionally, the stock's current drawdown of -25.7% from its recent peak is worse than -20%, indicating an overreaction selloff.*

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5. Patterns of Agreement and Disagreement

The results reveal a clear structural pattern: the two strategies disagree whenever a stock is in a directional state — moving strongly up or falling significantly — and agree only when a stock occupies a specific dual-signal zone.

The Momentum Trader's decision is almost entirely determined by two factors: whether price is above or below both moving averages, and whether the momentum score is positive or negative. These are measures of current direction. The Value Contrarian's decision is determined by two different factors: RSI level and distance from the 52-week high. These are measures of position relative to extremes. The two frameworks share no primary inputs.

This input separation explains why disagreement is the norm. A rising stock (positive momentum, above MAs) is almost always also near its 52-week high with elevated RSI — triggering Momentum BUY and Contrarian SELL simultaneously. A falling stock (negative momentum, below MAs) is almost always also moving away from its high with declining RSI — triggering Momentum SELL and Contrarian BUY simultaneously. The datasets confirm this: NVDA (rising), INTC (sharply rising), TSLA (falling), and PFE (declining) all produced opposite decisions.

The exception is GME, and it is analytically illuminating. GME is the only stock where the local trend (above both MAs, positive momentum) and the historical position (31% below 52-week high) point the same direction. The stock is recovering from a deep prior decline — which means momentum is locally positive but value is still available. This dual-signal structure is the only market condition that satisfies both philosophies simultaneously.

The MSFT case is subtle. The Momentum Trader issued a BUY because price was above both MAs, but the value of the divergence is that the Contrarian issued HOLD rather than BUY — demonstrating

that a 27% drawdown is not deep enough to trigger contrarian conviction. MSFT has not fallen far enough from its high to create the fear signal the Contrarian requires, even though it has fallen significantly in absolute terms. This is a useful calibration: the Contrarian only acts on extremes.

The market condition that drives the sharpest divergence is an explosive momentum surge — exactly what INTC exhibited. A +40% move in 30 days maximally satisfies Momentum's buy criteria while simultaneously tripping every Contrarian sell trigger. This is also the market condition most likely to produce extreme outcomes in reality: either a sustained breakout or a sharp reversal. The fact that both high-confidence agents disagree at confidence 9 and 8 respectively is itself informative — it signals genuine uncertainty about which framework is correct.

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6. Failure and Surprise Case

The NVDA Value Contrarian in the first run — prompt non-compliance

In the initial version of the prompts, the Value Contrarian called NVDA a BUY despite the stock having RSI of 68.07, being only 5.08% from its 52-week high, and gaining +10.78% in 7 days. The agent's stated justification was: "The RSI is at 68.07, indicating an oversold condition" — which is factually incorrect. RSI 68 is not oversold by any standard definition; oversold is conventionally RSI below 30, and the neutral range extends to approximately 65.

The root cause was prompt underspecification. The original prompt described the strategy's philosophy in qualitative terms ("buy fear, sell greed") but did not provide hard numeric thresholds. Without explicit cutoffs, the local model (llama3.2) defaulted to vague pattern-matching and produced justifications that sounded plausible but violated the agent's own stated rules.

The fix was to rewrite both prompts with explicit decision trees and numeric thresholds: RSI above 65 AND within 10% of 52-week high triggers SELL; RSI above 70 is always a SELL; RSI 45-65 is neutral, not oversold. After this change, the Value Contrarian correctly issued a SELL on NVDA in the second run.

This failure reveals an important limitation of smaller local models for structured decision-making tasks: they are susceptible to rationalization, finding post-hoc justifications for whatever decision they lean toward rather than faithfully applying stated criteria. The solution is not trust but constraint — the more precisely the prompt specifies numeric conditions, the less room the model has to deviate.

Secondary surprise — MSFT evaluator error: The MSFT evaluator output incorrectly stated that "the Momentum Trader and Value Contrarian have converged on a BUY recommendation" when the actual decisions were BUY and HOLD. The evaluator prompt instructs the model to note whether the strategies agree or disagree, but the model appears to have read both positive-leaning outputs as agreement. This is an evaluator hallucination — the model summarized the intent of the two justifications rather than the literal decision fields. The JSON correctly records agents_agree: false, but the analysis text contradicts it. For production use, the evaluator should be required to quote the literal decision values from the structured output rather than inferring them from the reasoning.

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7. Reflection

Which strategy to follow with $10,000 for one month?

Given the current market conditions reflected in this data — a mix of recovering large-caps, high-momentum tech, and declining names — the Value Contrarian is the more defensible choice for a one-month horizon. The Momentum Trader's strongest BUY signals are on INTC (up 40% in 30 days) and NVDA (near its 52-week high): both are stocks where the easy gains may already be priced in. Following momentum into a stock after a 40% surge carries significant reversal risk over a short horizon. The Contrarian's SELL on INTC and TSLA BUY, by contrast, are positioned to benefit from mean reversion — a mechanism more reliably active over one-to-four week periods than trend continuation.

A hybrid third strategy — the Trend-Aware Contrarian:

A hybrid strategy would combine the Value Contrarian's entry signals with the Momentum Trader's exit discipline. It would use the Contrarian's framework to identify entry points — deep drawdowns, low RSI, fear-driven selling — but apply the Momentum Trader's moving average crossovers to determine when to exit. This solves a known weakness of pure contrarian investing: catching a falling knife before the trend actually reverses. The hybrid would only enter a contrarian position once the 7-day momentum turns positive and price crosses back above the 20-day MA, using the momentum signal as confirmation that the value opportunity has begun to be recognized by the market. On this dataset, such a strategy might have produced a HOLD on PFE (value conditions met, but momentum not yet confirmed) rather than a premature BUY.

Appendix: AI Use Documentation

This appendix documents the use of AI assistance in building and analyzing this system, per the assignment's responsible use requirement.

A.1 Representative Prompts Used in Development

The following is the final Value Contrarian system prompt, after revision. This is stored in prompts/strategy_b.txt:

> *You are a Value Contrarian investor. Your philosophy: markets overreact in both directions. You buy genuine fear and sell genuine greed. SELL when RSI is above 65 AND the stock is within 10% of its 52-week high, OR when RSI is above 70. HOLD when RSI is between 40 and 65 and no extreme moves are present. BUY only when RSI is below 45 AND the stock is more than 20% below its 52-week high, OR when current drawdown is worse than -20% with RSI below 50. RSI 45-65 is NEUTRAL, not oversold.*

The final Momentum Trader prompt (strategy_a.txt) uses a parallel structure: explicit SELL triggers (price below both MAs, negative 30d change AND negative momentum), explicit HOLD conditions

(mixed MA signals, flat momentum), and BUY conditions requiring all three: above both MAs, positive recent change, and positive momentum score.

A.2 Output Excerpts

The following excerpts are taken directly from the JSON output files:

From INTC.json — the sharpest disagreement (confidence 9 vs. 8):

> *Momentum Trader: 'The stock is above both the 20-day MA ($50.05) and the 50-day MA ($47.88), indicating a bullish trend. The recent price change (30d) is positive (+40.24%) and the momentum score is +34.81, further supporting the buy signal.' [BUY, confidence 9]*
>
> *Value Contrarian: 'INTC is overbought with an RSI of 74.02, which is above the neutral range of 40-65. Additionally, its price is within 10% of its 52-week high, making it a sell signal.' [SELL, confidence 8]*

From TSLA.json — directionally opposite decisions:

> *Momentum Trader: 'Price is below both the 20-day MA ($367.34) and the 50-day MA ($391.7), indicating a bearish trend crossover signal. The recent 30-day change is negative (-9.70%) with a momentum score of -5.42.' [SELL, confidence 8]*
>
> *Value Contrarian: 'TSLA is oversold with an RSI of 45.88, currently 20% below its 52-week high of $489.88. The stock's current drawdown of -25.7% from its recent peak is worse than -20%, indicating an overreaction selloff.' [BUY, confidence 8]*

A.3 What Was Accepted, Revised, and Rejected

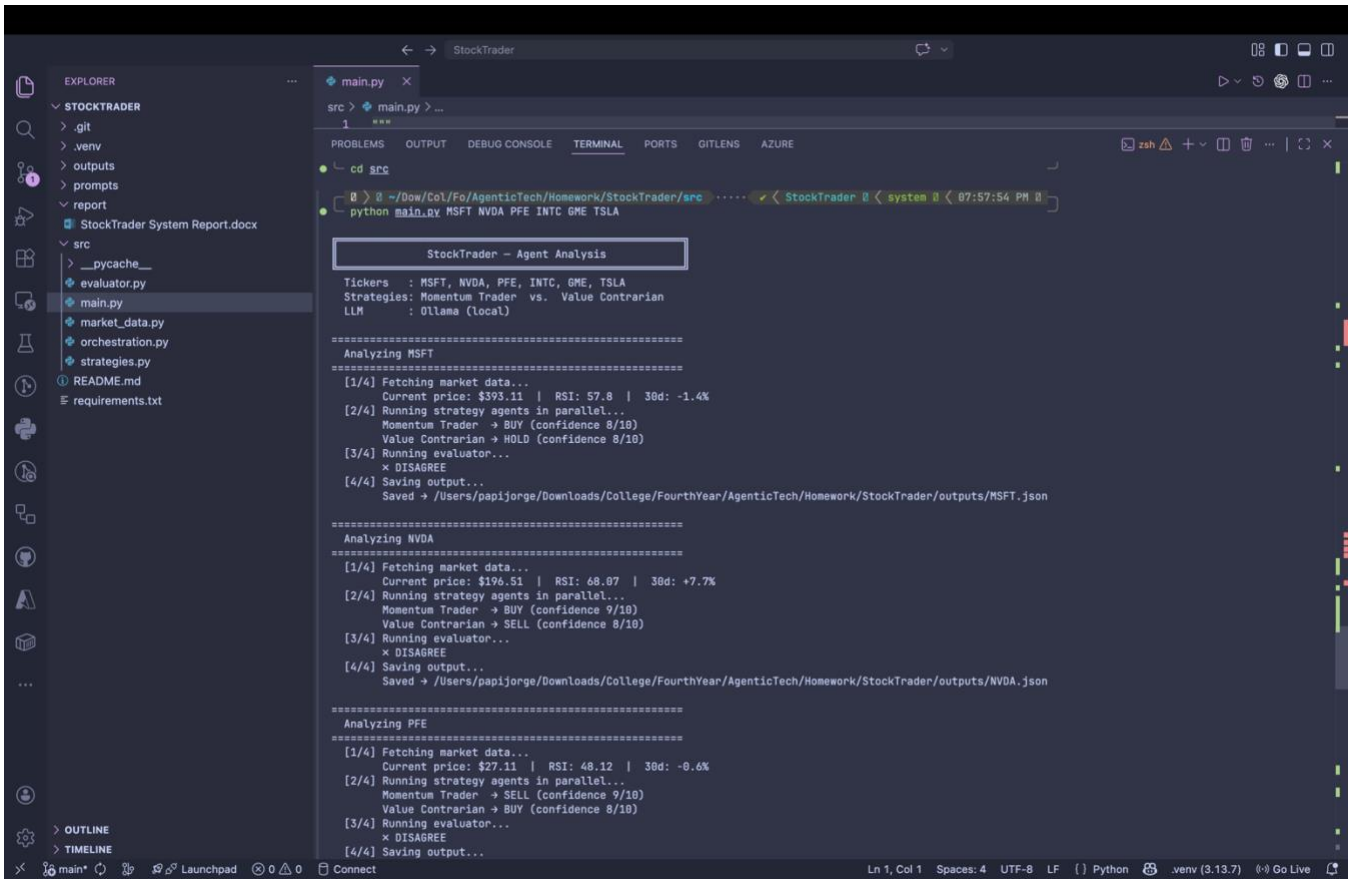
Claude (Anthropic) was used to generate the initial codebase including market_data.py, strategies.py, evaluator.py, orchestration.py, and main.py, as well as the initial prompt files. The architecture design — parallel execution via concurrent.futures, the indicator set, the JSON output structure — was accepted as generated.

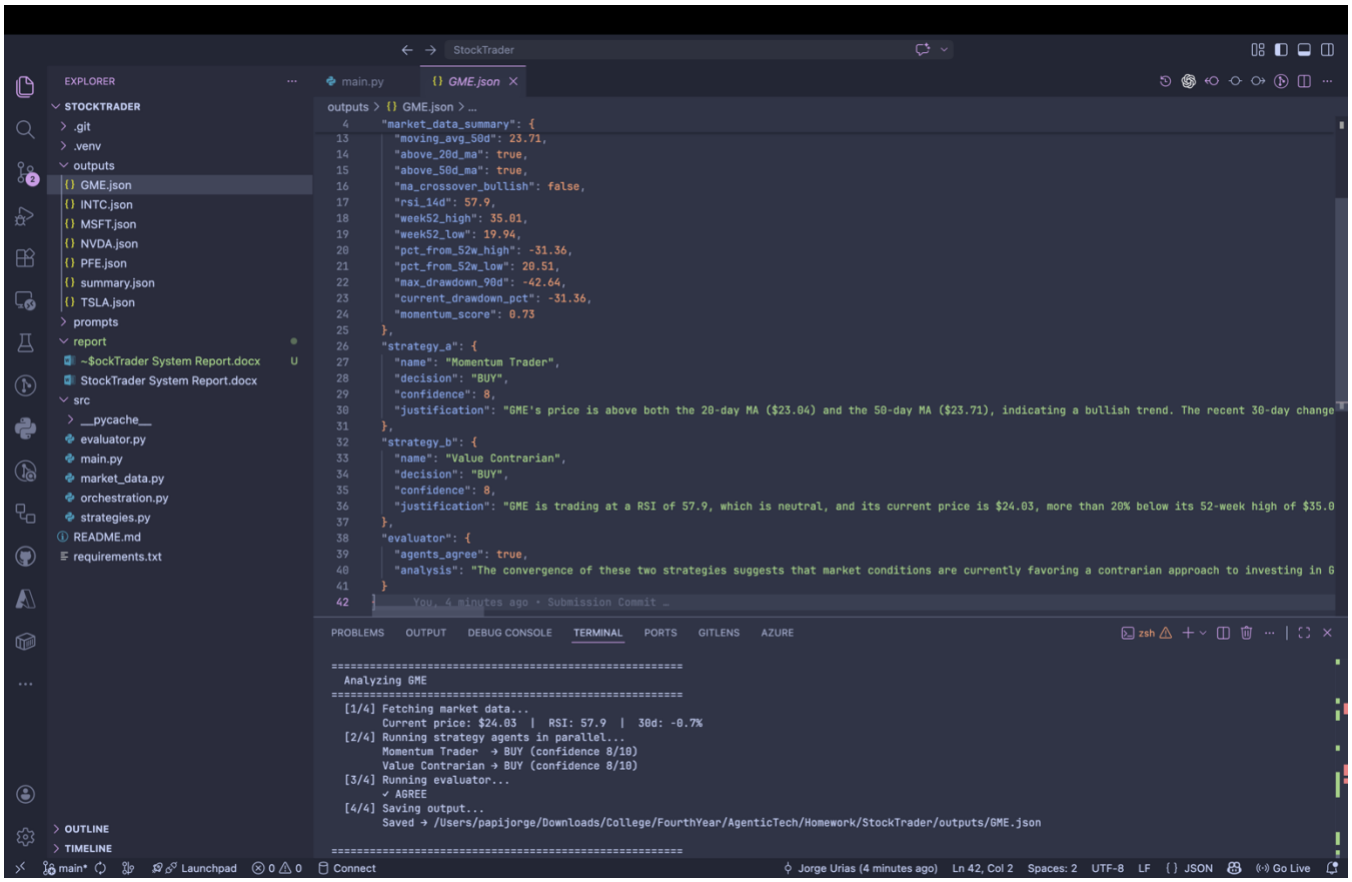
The initial prompts were revised after the first run revealed that the Value Contrarian was non-compliant, calling RSI 68 'oversold' and issuing BUYs on stocks near their 52-week highs. Both strategy prompts were rewritten with explicit numeric thresholds. The decision trees in the revised prompts (e.g., 'RSI above 70 is always a SELL') were authored based on standard technical analysis definitions, not AI-generated.

The analysis in this report — particularly the pattern identification in Section 5 and the hybrid strategy design in Section 7 — represents independent editorial judgment. The core insight that the two strategies share no primary inputs (MAs vs. RSI/52w-high) and that this input separation explains the pattern of disagreement was developed through examination of the JSON outputs, not accepted from AI output.

The MSFT evaluator error (Section 6, secondary failure) was identified by manual inspection of the output files and was not flagged by the code. It is documented here as an example of the editorial judgment required when working with LLM-generated outputs.

Outputs


StockTrader
EXPLORER
STOCKTRADER
.git
.venv
outputs
prompts
report
StockTrader System Report.docx
src
__pycache__
evaluator.py
main.py
market_data.py
orchestration.py
strategies.py
README.md
requirements.txt
OUTLINE
TIMELINE
main.py
src > main.py > ...
1 """
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS AZURE
cd src
~/Dow/Col/Fo/AgenticTech/Homework/StockTrader/src
StockTrader system 07:57:54 PM
python main.py MSFT NVDA PFE INTC GME TSLA
StockTrader — Agent Analysis
Tickers : MSFT, NVDA, PFE, INTC, GME, TSLA
Strategies: Momentum Trader vs. Value Contrarian
LLM : Ollama (local)
Analyzing MSFT
[1/4] Fetching market data...
Current price: $393.11 | RSI: 57.8 | 30d: -1.4%
[2/4] Running strategy agents in parallel...
Momentum Trader → BUY (confidence 8/10)
Value Contrarian → HOLD (confidence 8/10)
[3/4] Running evaluator...
× DISAGREE
[4/4] Saving output...
Saved → /Users/papijorge/Downloads/College/FourthYear/AgenticTech/Homework/StockTrader/outputs/MSFT.json
Analyzing NVDA
[1/4] Fetching market data...
Current price: $196.51 | RSI: 68.07 | 30d: +7.7%
[2/4] Running strategy agents in parallel...
Momentum Trader → BUY (confidence 9/10)
Value Contrarian → SELL (confidence 8/10)
[3/4] Running evaluator...
× DISAGREE
[4/4] Saving output...
Saved → /Users/papijorge/Downloads/College/FourthYear/AgenticTech/Homework/StockTrader/outputs/NVDA.json
Analyzing PFE
[1/4] Fetching market data...
Current price: $27.11 | RSI: 48.12 | 30d: -0.6%
[2/4] Running strategy agents in parallel...
Momentum Trader → SELL (confidence 9/10)
Value Contrarian → BUY (confidence 8/10)
[3/4] Running evaluator...
× DISAGREE
[4/4] Saving output...
main* Launchpad 0 0 Connect
Ln 1, Col 1 Spaces: 4 UTF-8 LF Python .venv (3.13.7) Go Live


StockTrader
EXPLORER
STOCKTRADER
.git
.venv
outputs
GME.json
INTC.json
MSFT.json
NVDA.json
PFE.json
summary.json
TSLA.json
prompts
report
~$ockTrader System Report.docx
StockTrader System Report.docx
src
__pycache__
evaluator.py
main.py
market_data.py
orchestration.py
strategies.py
README.md
requirements.txt
OUTLINE
TIMELINE
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GME.json
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},
"strategy_b": {
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},
"evaluator": {
"agents_agree": true,
"analysis": "The convergence of these two strategies suggests that market conditions are currently favoring a contrarian approach to investing in G
}
You, 4 minutes ago • Submission Commit …
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
GITLENS
AZURE
zsh
==========
Analyzing GME
==========
[1/4] Fetching market data...
Current price: $24.03 | RSI: 57.9 | 30d: -0.7%
[2/4] Running strategy agents in parallel...
Momentum Trader → BUY (confidence 8/10)
Value Contrarian → BUY (confidence 8/10)
[3/4] Running evaluator...
✓ AGREE
[4/4] Saving output...
Saved → /Users/papijorge/Downloads/College/FourthYear/AgenticTech/Homework/StockTrader/outputs/GME.json
==========
main*
Launchpad
0 0
Connect
Jorge Urias (4 minutes ago)
Ln 42, Col 2
Spaces: 2
UTF-8
LF
JSON
Go Live